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Roll No

EC-6003-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Antenna and Wave Propagation

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume suitable data if missing.

1. Write a note on the following:
 - i) Parabolic reflector antenna
 - ii) Microstrip antenna
2. Write about sky wave propagation.
3. Discuss Dolph-Chebyshev synthesis.
4. What are Binomial Arrays?
5. What is reciprocity theorem? Explain any one application.
6. An antenna has an effective height of 100 meters and the current at the base is 450 Amperes (r.m.s.) at 40,000 Hz. What is the power radiated? If the total resistance of the antenna circuit is 1.12 ohms, what is the efficiency of the antenna?

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7. Explain Babinet's principles and complementary antenna.
8. Calculate the transmission path distance for an ionospheric transmission that utilize a layer of height 200km. The angle of elevation of the antenna beam is 20° . The earth's radius can be assumed to be 6370km.

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